

1、 Suppose there is a loan system. Whether a person(Customer) can borrow money needs to meet the following three conditions: query whether he has enough deposits in the banking subsystem(Bank), query whether he has good credit in the credit subsystem(Credit), query whether he has bad loan records in the loan subsystem(Loan). The name of the facade class is Mortgage, and the name of client program is Client.

Bank

```
package experiment9_1;

class Bank {

    public boolean hasEnoughDeposits(Customer customer,
double amount){

        System.out.println("Checking bank for sufficient
savings...");

        return customer.getDeposits() >= amount;

    }

}
```

Client

```
package experiment9_1;

class Client {

    public static void main(String[] args) {

        Customer customer = new Customer("Tim", 45000,
1000, 0);

        Mortgage mortgage = new Mortgage();

        boolean flag = mortgage.isEligible(customer, 2000);

    }

}
```

```

        if(flag == true){
            System.out.println(customer.getName() + " is
eligible for the loan.");
        }else{
            System.out.println(customer.getName() + " is not
eligible for the loan.");
        }
    }
}

```

Credit

```

package experiment9_1;

class Credit {
    public boolean hasGoodCredit(Customer customer){
        System.out.println("Checking customer credit...");
        return customer.getcreditScore() >= 700;
    }
}

```

Customer

```

package experiment9_1;

class Customer {
    private String name;
}

```

```
private double deposits;

private int creditScore;

private int badLoanRecords;


public Customer(String name, double deposits, int
creditScore, int badLoanRecords){

    this.name = name;

    this.deposits = deposits;

    this.creditScore = creditScore;

    this.badLoanRecords = badLoanRecords;

}


public String getName(){

    return name;

}

public double getDeposits(){

    return deposits;

}

public int getcreditScore(){

    return creditScore;

}

public int getBadLoanRecords(){
```

```
        return badLoanRecords;
    }
}
```

Loan

```
package experiment9_1;

class Loan {
    public boolean hasBadLoanRecords(Customer customer){
        System.out.println("Checking loan history...");
        return customer.getBadLoanRecords() == 0;
    }
}
```

Mortgage

```
package experiment9_1;

class Mortgage {
    Bank enoughDeposits;
    Credit goodCredit;
    Loan badLoanRecords;

    public Mortgage(){
        enoughDeposits = new Bank();
        goodCredit = new Credit();
    }
}
```

```

        badLoanRecords = new Loan();
    }

    public boolean isEligible(Customer customer, double
loanAmount){
        System.out.println("Evaluating eligibility for " +
customer.getName() + "...");
        boolean con1 =
enoughDeposits.hasEnoughDeposits(customer, loanAmount);
        boolean con2 = goodCredit.hasGoodCredit(customer);
        boolean con3
= !badLoanRecords.hasBadLoanRecords(customer);
        return con1&&con2&&con3;
    }
}

```

Output

```

D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ IDEA 2
Evaluating eligibility for Tim...
Checking bank for sufficient savings...
Checking customer credit...
Checking loan history...
Tim is not eligible for the loan.

Process finished with exit code 0

```